Implementación de contenedores Docker y flujo de integración continua con GitHub Actions

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Integración Continua

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Docker-bulid y Docker compose up

```
PS C:\Users\JULIO\Documents\Proyectos\microservicios-py> docker compose up --build
 [+] Building 20.2s (16/16) FINISHED
  => [microservicios-py-api internal] load build definition from Dockerfile                                          0.1s
  => => transferring dockerfile: 274B                                                                                0.0s
  => [microservicios-py-api internal] load .dockerignore                                                             0.0s
  => => transferring context: 2B                                                                                     0.0s
  => [microservicios-py-store internal] load build definition from Dockerfile                                        0.0s
  => => transferring dockerfile: 274B                                                                                0.0s
  => [microservicios-py-api internal] load metadata for docker.io/library/python:3.12-slim                           1.8s
  => [microservicios-py-store internal] load .dockerignore                                                           0.0s
  => => transferring context: 2B                                                                                     0.0s
  => [microservicios-py-api internal] load build context                                                             0.1s
  => => transferring context: 1.45kB                                                                                 0.0s
  => [microservicios-py-store 1/5] FROM docker.io/library/python:3.12-slim@sha256:5e2bdb1b0d542d5039cffb8181c65930093949194  0.0s
  => [microservicios-py-store internal] load build context                                                           0.0s
  => => transferring context: 1.08kB                                                                                 0.0s
  => CACHED [microservicios-py-store 2/5] WORKDIR /srv                                                               0.0s
  => [microservicios-py-store 3/5] COPY requirements.txt .                                                           0.1s
  => [microservicios-py-api 3/5] COPY requirements.txt .                                                             0.1s
  => [microservicios-py-store 4/5] RUN pip install --no-cache-dir -r requirements.txt                               17.8s
  => [microservicios-py-api 4/5] RUN pip install --no-cache-dir -r requirements.txt                                 15.8s
  => [microservicios-py-api 5/5] COPY app ./app                                                                      0.1s
  => [microservicios-py-store] exporting to image                                                                    0.4s
  => => exporting layers                                                                                             0.1s
  => => writing image sha256:6db90e2f6031c7b6f041ea08048f69011bff57f3164832ebf9cd702378a182b9                        0.0s
  => => naming to docker.io/library/microservicios-py-api                                                            0.0s
  => => writing image sha256:50a51026ad92855fc5bb9ff77f176b98d42efe747a053fcfc9b7d06ccd8cf0d7                        0.0s
  => => naming to docker.io/library/microservicios-py-store                                                          0.0s
  => [microservicios-py-store 5/5] COPY app ./app                                                                    0.1s
 [+] Running 3/3
  - Network microservicios-py_backend  Created                                                                       0.3s
  - Container store                    Created                                                                       0.1s
  - Container api                      Created                                                                       0.1s
 Attaching to api, store
 store  |  * Serving Flask app 'main'
 store  |  * Debug mode: off
 store  | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
```

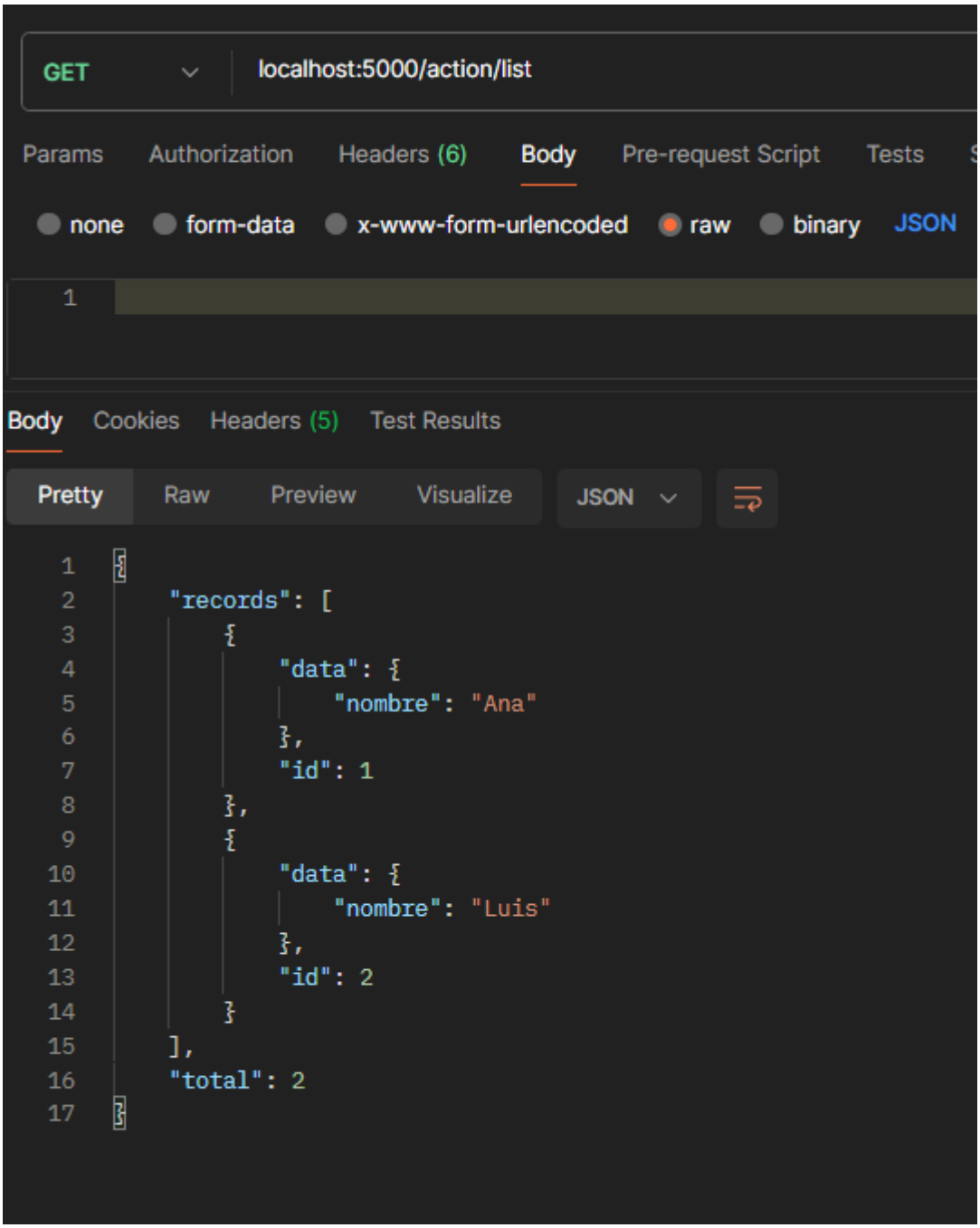
Docker ps

```
PS C:\Users\JULIO\Documents\Proyectos\microservicios-py> docker ps
CONTAINER ID   IMAGE                     COMMAND                  CREATED         STATUS                   PORTS                    NAMES
0a75369ad7d5   microservicios-py-api     "python -m app.main"     6 minutes ago   Up 6 minutes (healthy)   0.0.0.0:5000->5000/tcp   api
7dff63715a42   microservicios-py-store   "python -m app.main"     6 minutes ago   Up 6 minutes (healthy)   6000/tcp                 store
```

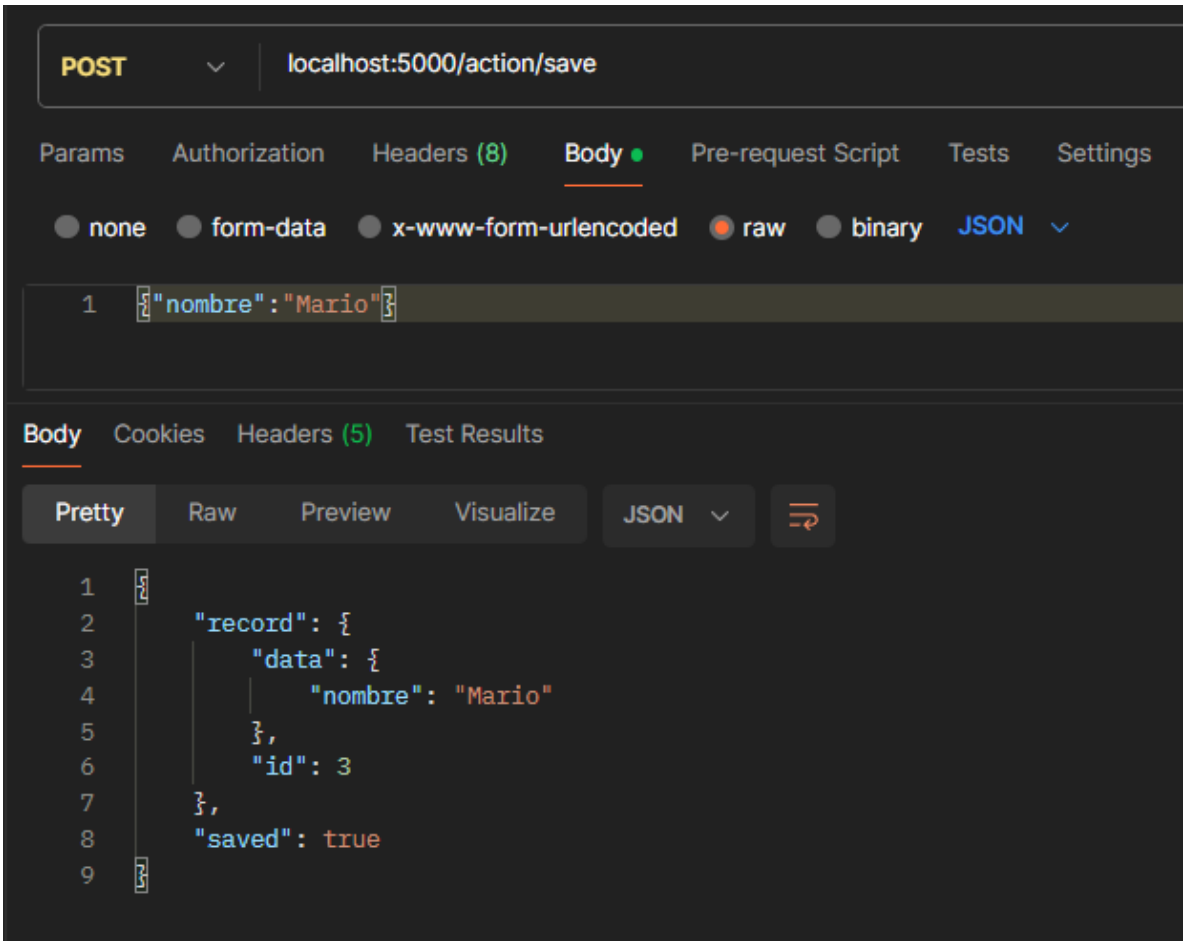
Docker network inspect

```
Display detailed information on one or more networks
PS C:\Users\JULIO\Documents\Proyectos\microservicios-py> docker network ls
NETWORK ID     NAME                                DRIVER    SCOPE
ef053551b821   bridge                              bridge    local
382ea549b63a   frontend_default                    bridge    local
180d1b7ed8cf   geco_multiplatform_geco_network     bridge    local
05c3655c4f3b   host                                host      local
10356e6958e7   microservicios-py_backend           bridge    local
9433017b69e2   none                                null      local
7d79e28e8704   red_postgresql                      bridge    local
PS C:\Users\JULIO\Documents\Proyectos\microservicios-py>
```

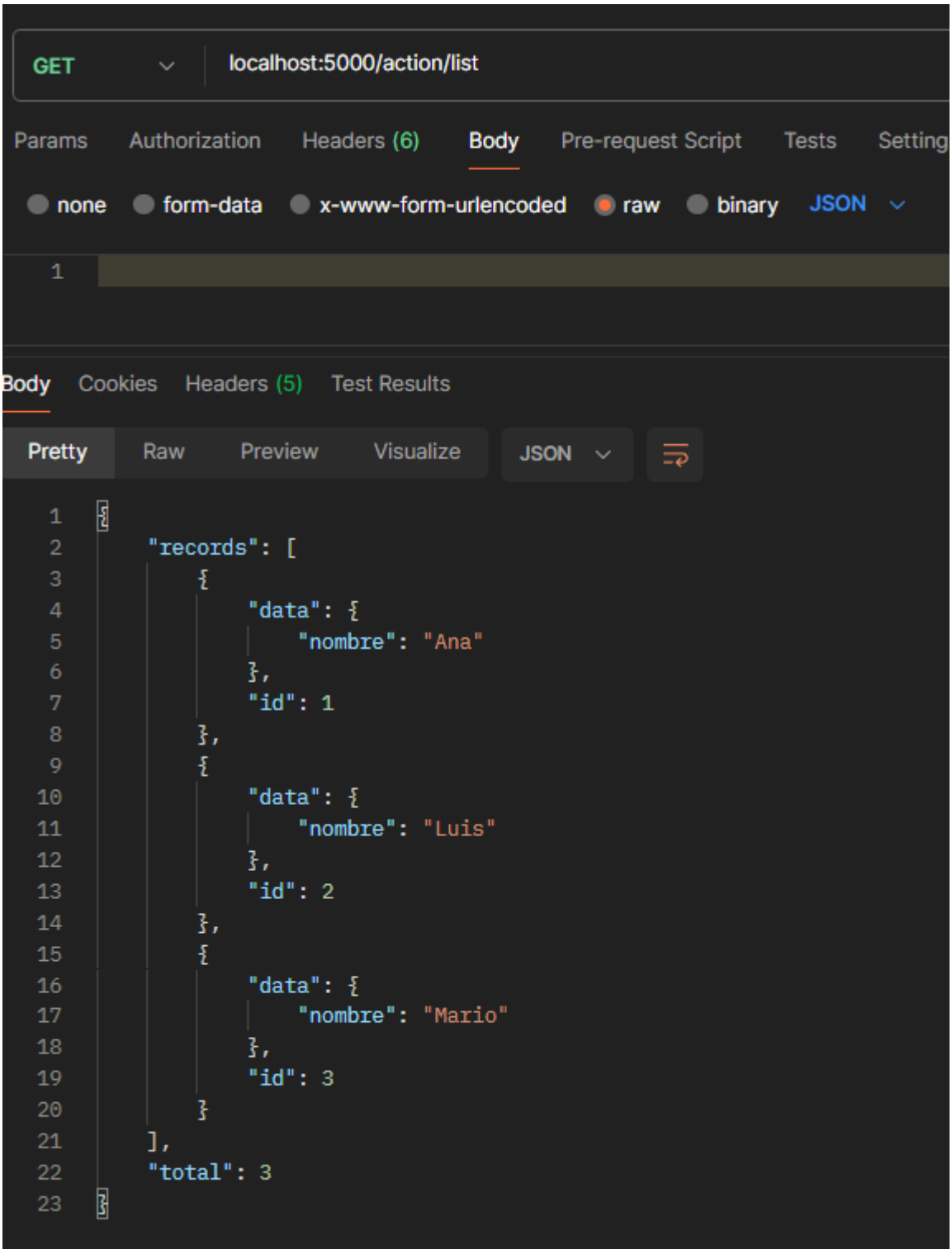
Endpoint vista



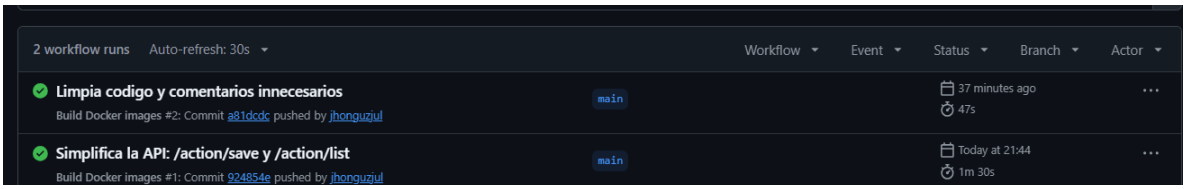
Endpoint para guardar



Verificación de conexión.


GET
localhost:5000/action/list
Params
Authorization
Headers (6)
Body
Pre-request Script
Tests
Setting
none
form-data
x-www-form-urlencoded
raw
binary
JSON
1
Body
Cookies
Headers (5)
Test Results
Pretty
Raw
Preview
Visualize
JSON
{
"records": [
{
"data": {
"nombre": "Ana"
},
"id": 1
},
{
"data": {
"nombre": "Luis"
},
"id": 2
},
{
"data": {
"nombre": "Mario"
},
"id": 3
}
],
"total": 3
}

Github actions



Diagrama



Conclusión

La implementación demuestra que Docker y Docker Compose permiten construir servicios reproducibles, comunicados por nombre de servicio dentro de una red propia según la necesidad. Git organiza el versionado en las ramas dev, qa y main que hay en este repositorio y GitHub Actions automatizó la construcción y prueba de las imágenes en cada push a main como flujo CI/CD asegurando una revisión continua según lo que se implemento.

Repositorio del proyecto: https://github.com/jhonguzjul/microservicios-py

Referencias

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